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Original Article

# Grooming and group cohesion in primates: implications for the evolution of language

Cyril C. Grueter [a,b,*], Annie Bissonnette [b,c], Karin Isler [b], Carel P. van Schaik [b]

[a] School of Anatomy, Physiology and Human Biology, The University of Western Australia, Crawley, Australia
[b] Anthropological Institute and Museum, University of Zurich, Switzerland
[c] Courant Research Centre Evolution of Social Behaviour, Research Group Social Evolution in Primates, Göttingen, Germany

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ABSTRACT

It is well established that allogrooming, which evolved for a hygienic function, has acquired an important derived social function in many primates. In particular, it has been postulated that grooming may play an essential role in group cohesion and that human language, as verbal grooming or gossip, evolved to maintain group cohesion in the hominin lineage with its unusually large group sizes. Here, we examine this group cohesion hypothesis and test it against the alternative grooming-need hypothesis which posits that rates of grooming are higher in species where grooming need (i.e. the motivation to groom for hygiene and its associated psychological reward) is more pronounced. This alternative predicts that the derived social function of grooming evolved mostly in those lineages that had the highest exposure to ectoparasites and dirt, i.e. terrestrial species. A detailed comparative analysis of 74 species of wild primates, controlling for phylogenetic non-independence, showed that terrestriality was a highly significant predictor of allogrooming time, consistent with the prediction. The predictions of the group cohesion hypothesis were not supported, however. Group size did not predict grooming time across primates, nor did it do so in separate intra-population analyses in 17 species. Thus, there is no comparative support for the group-cohesion function of allogrooming, which questions the role of grooming in the evolution of human language.

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## 1. Introduction

Allogrooming (also called social grooming) is a nearly universal element in the behavioral repertoire of many social mammals (Mooring, Blumstein, & Stoner, 2004; Kutsukake & Clutton-Brock, 2006), including non-human primates (Goosen, 1987) and humans (Nelson & Geher, 2007). It has been hypothesized to serve two main functions, a hygienic one (Hutchins & Barash, 1976; Barton, 1985; Zamma, 2002), and a social one (e.g. McKenna, 1978; Di Bitetti, 1997; Fedurek & Dunbar, 2009). Although the social function must be derived from the hygienic one and the two are not mutually exclusive (Saunders, 1988), the relative significance of these two functions is debated (e.g. Boccia, 1983).

There is good evidence for the hygienic function in nonhuman primates. Primates can become heavily burdened with ticks (Struhsaker, 1967; Freeland, 1981; Brain & Bohrmann, 1992), leeches (Wright et al., 2009), and lice (Boese, 1974), which can cause reduced fitness of their hosts, in terms of either survival or reproduction (Lehmann, 1993). Behavioral defenses of the host, such as grooming, are therefore thought to have evolved in response to these fitness costs (Hart, 1992). Grooming serves to remove dirt particles, defoliated skin and ectoparasites, such as lice and ticks, from the fur of conspecifics (Struhsaker, 1967). Consistent with a hygienic function of allogrooming, higher ectoparasite abundance elicits more grooming, and when grooming is prevented, ectoparasite abundance increases (Mooring, McKenzie, & Hart, 1996; Eckstein & Hart, 2000). Moreover, grooming activity targets body areas with the highest density of louse eggs (Zamma, 2002) as well as those parts that tend to remain ungroomed in autogrooming bouts (e.g. Hutchins & Barash, 1976; Barton, 1985; Saunders, 1988).

There is similarly convincing evidence that allogrooming serves social functions. The basic social function is to ensure the receipt of grooming by giving approximately equal amounts of grooming, as is often observed in many species, including primates (Barrett, Henzi, Weingrill, Lycett, & Hill, 1999; Leinfelder, de Vries, Deleu, & Nelissen, 2001; Schino & Aureli, 2010). Thus, it can be regarded as a direct correlate of the hygiene function. However, the high percentage of time devoted to grooming suggests that in many species allogrooming has also acquired a derived social function beyond the need for hygiene (Dunbar & Sharman, 1984). First, in some species allogrooming has short-term benefits, such as social tolerance at resources (e.g. Barrett, Gaynor, & Henzi, 2002; Henzi & Barrett, 2007; Port, Clough, & Kappeler, 2009). Moreover, in many species allogrooming is exchanged over a longer time frame for tolerance by dominant individuals or for delivery of food, sexual tolerance, sexual access or agonistic support (Schino, 2007; Schino, Polizzi di Sorrentino, & Tiddi, 2007; Schino & Aureli, 2008). Indeed, in many species dyadic allogrooming serves to establish, uphold, reinforce and improve social

* Corresponding author. The University of Western Australia, School of Anatomy, Physiology and Human Biology, 35 Stirling Highway, Crawley WA 6009, Australia.
*E-mail address:* cyril.grueter@uwa.edu.au (C.C. Grueter).

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**Table 1**
Summary table on group size and grooming (species means) as well as substrate and female body mass.

| Species | Group size (means of population means) (band/ community size in brackets) | Grooming (%) (means of population means) | Substrate | Female body mass (g) |
|---|---|---|---|---|
| *Avahi laniger* | 2 | 2 | A | 1316 |
| *Lemur catta* | 18.3 | 7.8 | T | 2210 |
| *Eulemur fulvus* | 9.7 | 8.3 | A | 1763 |
| *Propithecus verreauxi* | 5.6 | 2 | A | 2760 |
| *Saguinus mystax* | 4.0 | 1.85 | A | 538 |
| *Cebus olivaceus* | 16.5 | 1.9 | A | 2520 |
| *Cebus apella* | 14.6 | 1.21 | A | 2520 |
| *Cebus albifrons* | 10.75 | 0.615 | A | 2290 |
| *Callicebus torquatus* | 4 | 9.9 | A | 1210 |
| *Callicebus oenanthe* | 4 | 4.5 | A | 909[a] |
| *Chiropotes satanas* | 32 | 0.4 | A | 2960 |
| *Pithecia pithecia* | 8 | 0.8 | A | 1580 |
| *Cacajao calvus* | 44 | 1.8 | A | 2880 |
| *Alouatta palliata* | 13 | 0 | A | 5350 |
| *Alouatta seniculus* | 9.5 | 0.4 | A | 5210 |
| *Alouatta guariba* | 5.9 | 2.6 | A | 4450 |
| *Alouatta caraya* | 9.5 | 1.2 | A | 4330 |
| *Ateles geoffroyi* | 15 | 2.5 | A | 7700 |
| *Ateles belzebuth* | 16 | 0.1 | A | 8112 |
| *Brachyteles hypoxanthus* | 34 | 0 | A | 8070 |
| *Presbyis siamensis* | 18 | 0 | A | 6880 |
| *Presbytis potenziani* | 6 | 0.1 | A | 6404 |
| *Presbytis rubicunda* | 7 | 0 | A | 6170 |
| *Presbytis thomasi* | 5.8 | 1.3 | A | 6690 |
| *Trachypithecus leucocephalus* | 10.5 | 11.5 | T | 7820 |
| *Trachypithecus pileatus* | 13 | 0.4 | A | 9860 |
| *Trachypithecus francoisi* | 9.5 | 1.2 | T | 7350 |
| *Semnopithecus achates* | 12 | 6 | T | 11700 |
| *Semnopithecus entellus* | 20 | 6 | T | 9890 |
| *Semnopithecus* cf. *schistaceus* | 63.8 | 8.4 | T | 14800 |
| *Rhinopithecus bieti* | 11.5 (293) | 6.7 | T | 9100 |
| *Rhinopithecus avunculus* | 13.3 | 5.6 | A | 8300 |
| *Nasalis larvatus* | 20 | 2.8 | A | 9820 |
| *Pygathrix nigripes* | 11.6 | 2.25 | A | 8700 |
| *Pygathrix nemaeus* | 28 | 1.78 | A | 8440 |
| *Colobus guereza* | 9 | 6.2 | A | 7900 |
| *Colobus satanas* | 16 | 5.5 | A | 7420 |
| *Colobus polykomos* | 9 | 0.7 | A | 8300 |
| *Colobus angolensis* | 300 | 5 | A | 7570 |
| *Colobus vellerosus* | 18.3 | 1 | A | 6900 |
| *Piliocolobus rufomitratus* | 19 | 2.1 | A | 7214 |
| *Piliocolobus badius* | 27 | 5.4 | A | 8210 |
| *Piliocolobus kirkii* | 34.6 | 7 | A | 5460 |
| *Piliocolobus tephrosceles* | 60 | 5.2 | A | 6728 |
| *Procolobus verus* | 3 | 3.6 | A | 4200 |
| *Cercopithecus mitis* | 26.9 | 8 | A | 3930 |
| *Cercopithecus ascanius* | 32.5 | 5.6 | A | 2920 |
| *Cercopithecus campbelli* | 9 | 2.8 | A | 2700 |
| *Cercopithecus diana* | 28.8 | 2.5 | A | 3900 |
| *Chlorocebus aethiops* | 12.4 | 6.8 | T | 2980 |
| *Chlorocebus djamdjamensis* | 53 | 2.7 | A | 2980[b] |
| *Lophocebus albigena* | 15 | 5.2 | A | 6010 |
| *Cercocebus galeritus* | 27 | 5.5 | A | 5260 |
| *Erythrocebus patas* | 28.7 | 4.53 | T | 6500 |
| *Macaca fuscata* | 27.7 | 18 | T | 8030 |
| *Macaca mulatta* | 84 | 6.6 | T | 5367 |
| *Macaca munzala* | 18 | 12 | T | 10000[c] |
| *Macaca fascicularis* | 29.3 | 10.6 | A | 3590 |
| *Theropithecus gelada* | 13 (144.7) | 17.4 | T | 11700 |
| *Papio anubis* | 58.8 | 7.1 | T | 13300 |
| *Papio papio* | 192.5 | 8.3 | T | 12000 |
| *Papio hamadryas* | 7.5 (51) | 13.5 | T | 9900 |
| *Papio ursinus* | 29.3 | 12.8 | T | 14800 |
| *Hylobates lar* | 4.6 | 4.4 | A | 5340 |
| *Hylobates klossii* | 3.8 | 0 | A | 5920 |
| *Hylobates agilis* | 4 | 0 | A | 5820 |
| *Hylobates pileatus* | 6 | 5 | A | 5440 |
| *Symphalangus syndactylus* | 4.3 | 10 | A | 10710 |
| *Pan troglodytes* | 10 (63) | 9 | T | 40367 |
| *Pan paniscus* | 6.2 (27.8) | 5.7 | T | 33200 |
| *Gorilla gorilla* | 11 | 0.1 | T | 71500 |
| *Gorilla beringei* | 17.6 | 2.5 | T | 97500 |
| *Pongo pygmaeus* | 1.2 | 0.01 | A | 35800 |
| *Pongo abelii* | 1.8 | 0.01 | A | 35600 |

A = mainly arboreal, T = terrestrial or semi-terrestrial. References for grooming and group size are in Table S1 (available on the journal's website at www.ehbonline.org). References for substrate were taken from Plavcan and van Schaik (1997) and Rowe and Myers (2011); additional data on terrestriality are from Grueter et al. (in press-b), Huang and Li (2005), Lehmann et al. (2007). Body size data are mainly from Smith and Jungers (1997), complemented with data from Delson, Terranova, Jungers, Sargis, Jablonski, and Dechow (2000), Di Fiore and Campbell (2007), Ford and Davis (1992), Gordon (2006), Huang and Li (2005), Isler, Kirk, Miller, Albrecht, Gelvin, and Martin (2008), Kirkpatrick (1998), Mishra and Sinha (2008), Moore (1985), Richard, Dewar, Schwartz, and Ratsirarson (2000), Rowe and Myers (2011).

[a] For *Callicebus oeanthe*, we used the female body mass of the closely related *C. donacophilus*.

[b] For *Chlorocebus djamdjamensis*, the body mass value was estimated to be identical to *C. aethiops*.

[c] Female value derived from sexual dimorphism of *M. thibetana*.

bonds among individuals. Finally, grooming may also play an essential role in group cohesion (Dunbar, 1991; Lehmann, Korstjens, & Dunbar, 2007; see below).

Grooming, be it for hygienic or for social benefits, takes up time. Allogrooming time varies widely among species (Dunbar, 1991; Lehmann et al., 2007), being virtually absent in some species of New World monkeys and colobines but occupying more than 10% of the activity budget in some baboons and macaques (Table 1). The aim of this paper is to identify the determinants of the variation in allogrooming time in primate groups, i.e. the time the average individual spends allogrooming. We examine two broad hypotheses: the *grooming-need hypothesis*, which reflects the original hygiene function of grooming, and the *group-cohesion hypothesis*, which represents one of the derived social functions of grooming with a predicted effect on a group member's mean allogrooming time.

We will now first elaborate the grooming-need hypothesis. The major determinant of grooming need is thought to be substrate use. Animal traffic is densest on the ground. This is obvious in two-dimensional environments such as savannas, but even in forests mammal biomass tends to be largest on the ground (Karanth & Sunquist, 1992), the more so the denser the understory (Emmons, 1984). As a result, densities of ticks and lice are expected to be highest near the ground. Indeed, a comparison of patterns of parasitism between terrestrial ring-tailed lemurs and arboreal sifakas has shown that lemur terrestriality is associated both with a higher number and a wider diversity of ectoparasites (Loudon, 2009). In addition, terrestrial animals move longer distances each day than arboreal ones (Clutton-Brock & Harvey, 1977), increasing the probability of contracting ectoparasites. Finally, dirt particles, and accordingly skin

irritation, are much more common on or near the ground than in the canopy, not only in savannas but also in forests (Ungar, 1994). These factors combined should lead to higher grooming need for terrestrial than arboreal primates. Moreover, it has been shown for ungulates that larger species need less grooming time (Mooring et al., 2004), so it seems reasonable to expect an effect of body size in primates as well, and we therefore include it in the statistical model.

Grooming has acquired social functions as well. Behavioral biologists would expect that fitness-enhancing activities are accompanied by a high motivation to perform them (Tinbergen, 1963), which in turn requires some physiological reward system. In the case of grooming, since most mammals, including primates, cannot reach all parts of their body with hands or mouth, this motivation should extend to grooming others in approximately equal amounts. Accordingly, both active allogrooming and passive allogrooming are associated with a release of endorphins (Keverne, Martensz, & Tuite, 1989; Dunbar, 2010), which act as a psycho-pharmacological reward. Moreover, we should expect allogrooming to reduce an animal's motivation to move on, so as not to curtail valuable grooming bouts. Accordingly, grooming tends to reduce heart rates and relax the groomees (Boccia, Reite, & Laudenslager, 1989; Aureli, Preston, & de Waal, 1999) and the groomers (Shutt, MacLarnon, Heistermann, & Semple, 2007).

This proximate regulatory system facilitated the evolution of the basic social allogrooming function for hygienic purposes, but also allowed for its cooptation for derived, "strategic" social purposes (e.g. grooming for tolerance). This process most likely occurred where grooming need (and thus the motivation to perform grooming) was high. As a result, we expect that the highest grooming time, be it for hygienic or derived, social purposes, should correlate with high intrinsic grooming need.

We now turn to the group-cohesion hypothesis, which is based on the fact that one of the derived social functions, group cohesion, is expected to have an effect on mean grooming time in a group. Dunbar (1991) found a correlation between time spent grooming and group size in a sample of primate species (although not in Platyrrhini [New World Monkeys]). Based on this finding, he suggested that an increasing group size imposes additional social challenges upon the animals. As a consequence, animals must spend an increasing proportion of their time allogrooming, in order to maintain the cohesion of their social group. They can do so by allocating an increasing amount of grooming to each member of the group equally or increase grooming towards valuable alliances partners (Dunbar, 1991). This argument was then extended to hominins. Based on a tight correlation between group size and neocortex ratio (Dunbar, 1993), Aiello and Dunbar (1993) estimated group sizes of *Homo sapiens* to be in the range of 150. To maintain the cohesion of a group of this size, however, individuals would have to spend over 40% of their time engaged in allogrooming, which is ecologically impossible. Natural selection would thus have favored alternative ways of fulfilling the function of maintaining group cohesion in the hominin lineage with its unusually large group sizes. Dunbar (1993, 1996) proposed that language, in the form of verbal grooming, fulfilled this vital function, because it is more compatible with other activities and can be directed at more than one other group member simultaneously. This hypothesis for the evolution of human language has become very influential (Gamble, 1998; Huron, 2001; Falk, 2004; McComb & Semple, 2005). Because the group-size effect on allogrooming is critical to this hypothesis, it is highly relevant to test its generality for primates.

Two approaches to test the robustness of the group-size effect can be recognized. First, Lehmann et al. (2007) corroborated Dunbar's earlier correlation between grooming and group size by demonstrating an asymptotic increase in time spent grooming with group size, but their sample excluded Platyrrhini. In contrast, Majolo, de BortoliVizioli, and Schino (2008) did not find statistical evidence that time spent on grooming is affected by group size in primates in a meta-analysis of intraspecific studies. The second approach consists of examining the relationship within populations of a single species. This offers a powerful test, because it eliminates potential confounding variables such as body size and habitat that interspecific analyses cannot control for. The few studies that explicitly provided intraspecific analyses also yielded equivocal results (Henzi, Lycett, & Weingrill, 1997; Sanchez-Villagra, Pope, & Salas, 1998; Teichroeb, Saj, Paterson, & Sicotte, 2003; Majolo et al., 2009).

Given these ambivalent results on the group cohesion hypothesis, the aim of this study is to thoroughly evaluate the group-size effect on allogrooming in primates, and to test it against and alongside the alternative grooming-need model. The two hypotheses make different predictions. The grooming-need hypothesis predicts that the factors that affect the importance of the hygiene function also affect the percentage of time spent allogrooming in a species. It thus expects clear effects of terrestriality and body size on allogrooming time (Model 1). The group-cohesion model, in contrast, makes one strong prediction, namely that allogrooming time is positively correlated with group size, both across and within species, regardless of lineage (Model 2). The grooming-need hypothesis and the group-cohesion hypothesis are not exclusive. The combined application of the two hypotheses would predict that the group-cohesion effect of allogrooming is highest in those species where grooming could acquire a strategic social function due to high grooming need. Thus, this combined model predicts a strong interaction effect between terrestriality and group size and one between body size and group size (Model 3).

## 2. Materials and methods

All data compiled for this study were taken from the literature on primate behavior (Tables 1, S1, available on the journal's website at www.ehbonline.org). Data on mountain gorillas were gathered by Dian Fossey Gorilla Fund International through its Karisoke Research Center. Allogrooming times were taken as percentage of total time, usually estimated through scan sampling. Mean allogrooming time

**Table 2**
Phylogenetic general least-squares model with grooming time as the response variable, and terrestriality, body mass and group size and interactions between them as possible effects, using unit or party size as a measure of group size for species with fission–fusion and modularity.

| | Primates | | Catarrhini | | Cercopithecoidea | | Platyrrhini | |
|---|---|---|---|---|---|---|---|---|
| N | 71 | | 54 | | 43 | | 16 | |
| $R^2$ adjusted | 0.230 | | 0.363 | | 0.208 | | 0.323 | |
| AICc | 440.9 | | 338.9 | | 262.5 | | 95.2 | |
| Lambda: ML-estimate | 0.428 | | 0 | | 0.781 | | 0 | |
| p of lambda >0 | 0.316 | | 1 | | 0.065 | | 1 | |
| p of lambda <1 | <0.0001 | | <0.0001 | | <0.0001 | | 0.0001 | |
| | *p* | *effect* | *p* | *effect* | *p* | *effect* | *p* | *effect* |
| ln female body mass | 0.043 | −2.02 | 0.032 | −2.48 | 0.704 | 0.89 | 0.213 | −2.03 |
| terrestriality | <0.0001 | 7.18 | <0.001 | 7.54 | 0.006 | 5.50 | all arboreal | |
| ln group size | 0.250 | 0.79 | 0.049 | 1.49 | 0.734 | 0.26 | 0.189 | −2.28 |

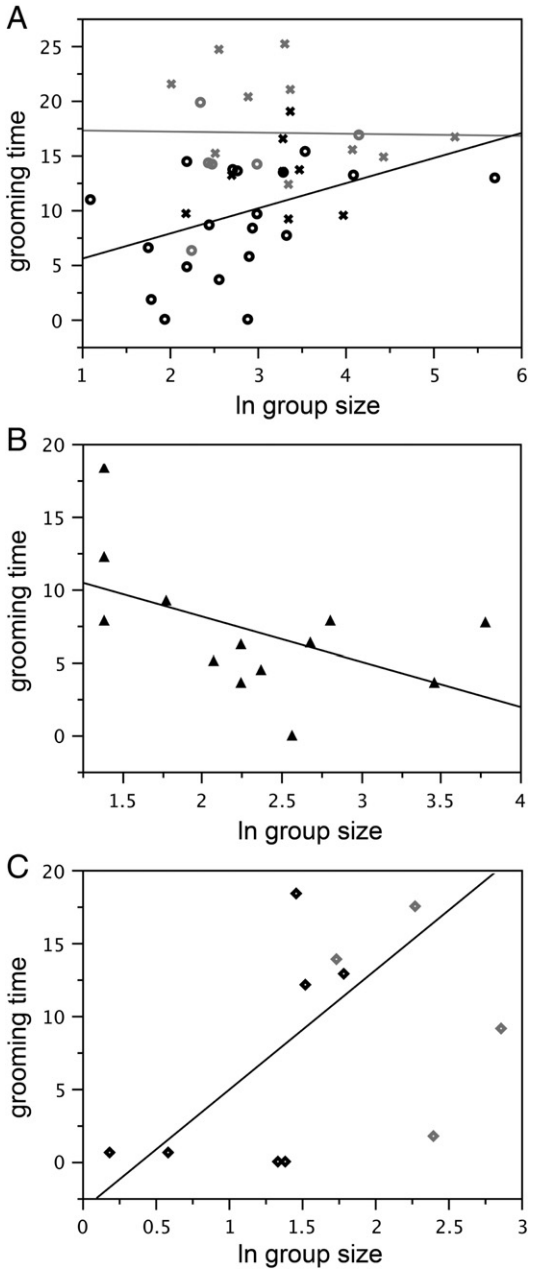



**Fig. 1.** Percentage of time spent grooming (angular transformed) vs. ln group size (unit/party size) in three taxonomic groups of primates. Black symbols: arboreal species, grey symbols: terrestrial species. a) Cercopithecoidea (n = 43). Colobinae (circles), Cercopithecinae (crosses). Arboreal species: n = 27, $r^2 = 0.171$, p-value of slope = 0.032, beta = 2.296. Terrestrial species: n = 16, $r^2 = 0.0003$, p-value of slope = 0.946, beta = −0.098. b) Platyrrhini (triangles), all arboreal: n = 13 species, $r^2 = 0.268$, p-value of slope = 0.070, beta = −3.095. c) Hominoidea (rhombs). Arboreal species: n = 7, $r^2 = 0.361$, p-value of slope = 0.154, beta = 8.199. Terrestrial species: n = 4.

was calculated for each study site, and in case of multiple study sites for a species, the values from different sites were averaged. Criteria for data inclusion were as follows. First, the populations had to be wild and unprovisioned, because provisioning, like captivity, tends to increase time spent allogrooming (e.g. Marriott, 1988). Second, if a population was represented by two studies conducted at different times, we use only one study to avoid pseudoreplication, and selected the study with the longer duration or the larger sample size. There was one exception: the same study group of siamangs was observed by different authors with strikingly varying results, but since no observation hours were given by Gittins and Raemaekers (1980), we used the mean of both studies for the analysis. Third, we only included data when grooming time estimates and group size were available from the same groups. Fourth, we only included grooming data from a study if grooming was recorded as a separate activity category, and not when grooming was part of a broader category such as "social" or "miscellaneous/other".

The main independent variables were group size, body size and terrestriality. We calculated mean group size per study site, and in case of multiple study sites averaged the values from different sites. Group size is unambiguous in most primates, but in species with fission–fusion dynamics (chimpanzees, bonobos, spider monkeys) and modularity (hamadryas, geladas, snub-nosed langurs), it can refer to either party size/one-male unit size or total size of the community/band (cf. Aureli et al., 2008; Grueter et al., in press-a). In these cases, we ran the analyses once for *actually* available grooming partners (party size in species with fission–fusion, one-male unit size in modular species) and once for *potentially* available grooming partners (community in species with fission–fusion and band size in species with modularity).

Body mass data were compiled from different sources (Table 1). We used female body mass to avoid an impact of sexual dimorphism, because females usually make up the majority of group members. Terrestriality was scored as a binary variable. A species was considered terrestrial if it spends more than 20% of its active period on the ground (Table 1).

For the comparative analysis of allogrooming time in different species, we built multivariate linear regression models to simultaneously assess the effect of several predictor variables on the dependent variable (percentage of grooming in activity budget). The predictions of the two hypotheses were tested with a statistical model including grooming time as the response variable, terrestriality, body mass and group size, and the interactions between these variables as possible effects.

To satisfy assumptions of normality, group size was natural log transformed and percentage of grooming was arcsine square root (angular) transformed. The transformed values of group size and grooming were normally distributed (Shapiro–Wilks: ln group size: W = 0.982, p = 0.393; grooming: W = 0.945, p = 0.159). Terrestriality was, for the phylogenetic analyses, treated as pseudo-continuous variable (1: terrestrial/0: arboreal). Phylogenetic analyses were done with PGLS (package CAIC (Orme, Freckleton, Thomas, Petzoldt, & Fritz, 2009) in R (R Development Core Team, 2010). The use of a phylogenetic regression was considered necessary because the maximum likelihood estimates of lambda (function pglmEstLambda in CAIC) were mostly larger than 0.5. However, lambda values were often not significantly different from zero. This combination of relatively high values of lambda with a high degree of uncertainty of the estimate may indicate that the amount of phylogenetic signal in the data varies between subgroups. The tree was taken from Perelman et al. (2011), complemented with the 10kTrees consensus tree version 2 (Arnold, Matthews, & Nunn, 2010) and estimated positions for some recently split species.

Intra-specific correlations were done in R using the Spearman method. Statistics were considered significant at $p < 0.05$.

## 3. Results

Interactions (group size × terrestriality and group size × body mass) had no significant effect on the response variable grooming time and were thus omitted from the models (Table S2 for unit size, Table S3a, available on the journal's website at www.ehbonline.org for band size). Terrestriality was always highly positively correlated with grooming time, irrespective of whether the analyses concerned all primates together or certain taxonomic groups and irrespective of whether community/band size or party/unit size was used (Tables 2, S3b, available on the journal's website at www.ehbonline.org). Within

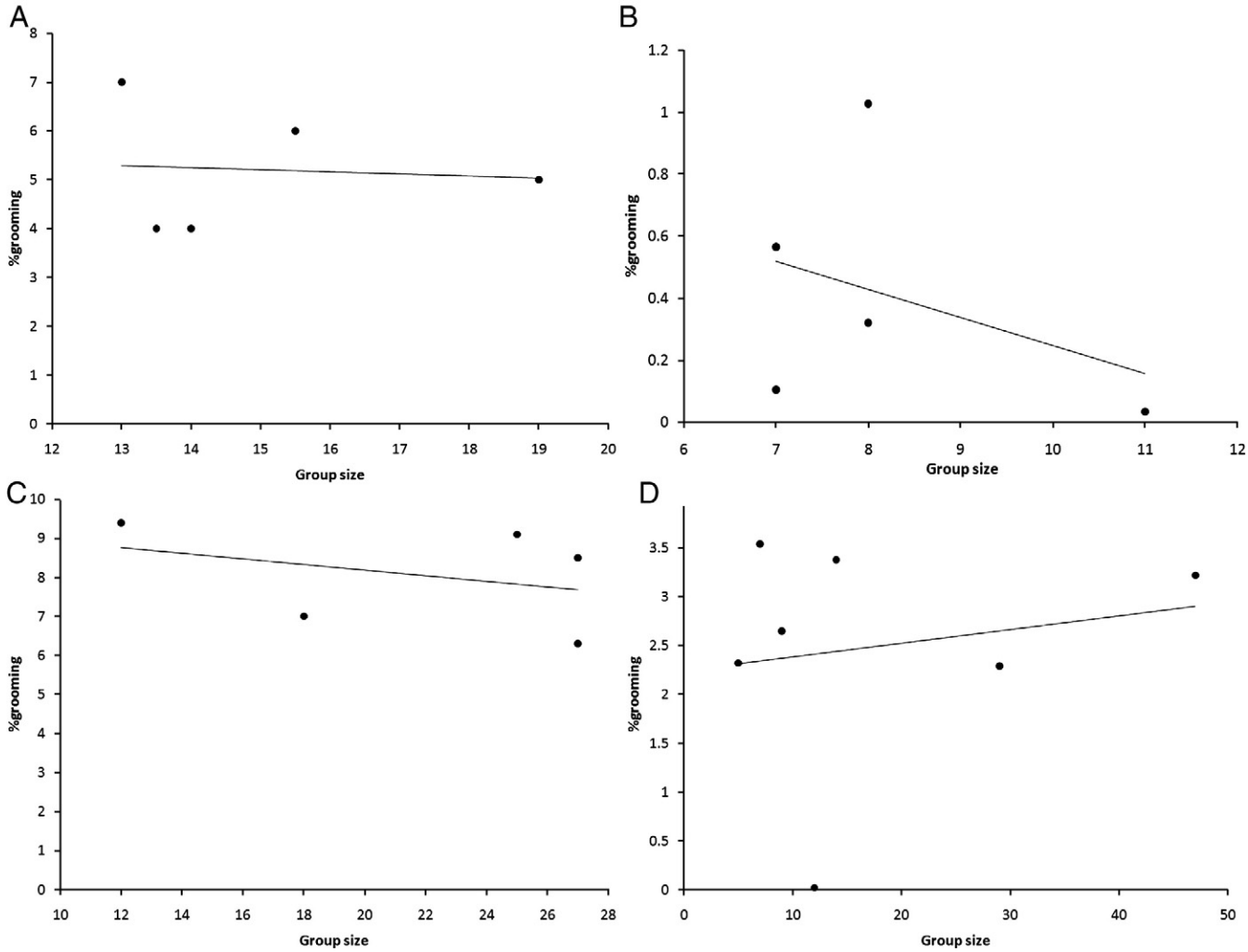



**Fig. 2.** Intra-population (between-group) correlations between group size and time spent grooming for species with at least 5 data points. a) *Lophocebus albigena*: $r_s = -0.154$, $p = 0.805$ (Chancellor & Isbell, 2009); b) *Alouatta seniculus*: $r_s = -0.316$, $p = 0.604$ (Sanchez-Villagra et al., 1998); c) *Cercopithecus mitis*: $r_s = -0.616$, $p = 0.269$ (Butynski, 1990); d) *Gorilla beringei*: $r_s = -0.071$, $p = 0.906$ (Karisoke Research Center, unpubl.). In the case of *Gorilla beringei*, the slope is so close to zero that a Spearman rank correlation produces a different sign from a Pearson correlation.

primates as a group and within Catarrhini, body mass was significantly negatively correlated with grooming time, but not within Cercopithecoidea and Platyrrhini (Tables 2, S3b, available on the journal's website at www.ehbonline.org). Group size did not have a significant effect on grooming time within primates as a group when taking into account *actual* grooming partners (party/unit size) ($p = 0.250$), but it was borderline significant when expanding it to *potential* grooming partners (community/band size) ($p = 0.045$). However, a positive effect of group size was found only within catarrhines ($p = 0.049$ for unit size, $p = 0.005$ for band size), but neither within Cercopithecoidea nor within Platyrrhini (Fig. 1). Thus, the positive effect of group size within the Catarrhini is entirely due to the positive

**Table 3**
Intra-specific (intra-population) comparisons between group size and grooming time for species with less than 5 data points.

| Taxonomic group | Species | Site | n (grps) | Sign of relationship | Source |
|---|---|---|---|---|---|
| Platyrrhini | *Cebus olivaceus* | Hato Masaguaral | 2 | + | deRuiter (1986) |
| | *Cebus albifrons* | Manu | 2 | - | van Schaik, unpubl. |
| | *Alouatta guariba* | El Piñalito | 2 | - | Agostini, unpubl. |
| | *Alouatta caraya* | El Piñalito | 2 | + | Agostini, unpubl. |
| Catarrhini–Cercopithecoidea | *Colobus guereza* | Kakamega | 2 | - | Fashing (2001) |
| | *Colobus vellerosus* | Boabeng-Fiema | 3 | + | Teichroeb et al. (2003) |
| | *Piliocolobus kirkii* | Jozani-Forest groups | 3 | - | Siex (2003) |
| | *Piliocolobus kirkii* | Jozani-Shamba groups | 4 | + | Siex (2003) |
| | *Macaca fuscata* | Yakushima | 2 | + | Agetsuma (1995), Maruhashi (1981) |
| | *Macaca fascicularis* | Ketambe | 3 | - | van Schaik, unpubl. |
| | *Papio ursinus* | Drakensberg | 2 | + | Barrett et al. (1999) |
| | *Papio ursinus* | De Hoop | 2 | + | Barrett et al. (1999) |
| | *Papio anubis* | Gilgil | 2 | - | Eley, Strum, Muchemi, and Reid (1989) |
| | *Erythrocebus patas* | Mutara | 2 | - | Chism and Rogers (2002) |
| Catarrhini–Hominoidea | *Hylobates lar* | Kao Yai | 2 | + | Bartlett (1999) |

relationship within Hominoidea, and the difference between Cercopithecoidea and Hominoidea.

For 4 species, there were data available for five or more groups of the same population (thus holding body size and habitat constant). In all 4 species there was a negative correlation between group size and grooming time (Fig. 2). For another 15 populations(13 species), we had data from 2 to 4 groups of the same population. Small sample sizes prohibited formal correlation analysis, but in 7 of the 15 populations, the relationship between group size and grooming was negative (Table 3). There was also no taxonomic effect on the sign of the correlation, in that neither Catarrhini nor Platyrrhini showed consistent effects. Thus, both the larger and smaller intraspecific samples indicate no consistent effect of group size on time spent allogrooming.

## 4. Discussion

Neither the inter- nor the intra-specific results support the group-cohesion model because there was no systematic relationship between grooming time and group size. Instead, they support the grooming-need model because terrestriality, the proxy used here for grooming need, explained variation in grooming time. However, the grooming-need hypothesis does not predict that allogrooming is solely concerned with hygiene. It is consistent with both the basic social function to ensure receiving enough grooming for the hygienic function and the derived social functions.

To see why this is, consider Fig. 3. There are at least 4 scenarios that one can envisage for how grooming has been co-opted to serve social goals. Scenarios a–c account for the finding of a strong effect of grooming need on observed grooming time in primates. In scenario a, the time allocated to the strategic use of allogrooming increases proportionally to the original grooming need imposed by hygienic considerations; in scenario b, a "grooming need threshold" needs first to be reached before grooming can be co-opted for other social functions; in scenario c, the cooptation of the basic hygiene function for derived, strategic purposes is strongest in species where grooming need is highest. The only scenario that would not account for our findings is scenario d.

In scenarios a–c, grooming can be pressed into service for an entirely different functional system (cf. Dunbar, 1991) to serve derived social functions provided grooming need (higher exposure to ectoparasites) and thus the motivation to perform grooming is high enough. The derived social use may largely be about gaining social tolerance (grooming calms the groomee down), which is in agreement with the fact that grooming roles tend to be biased, with subordinates grooming more than dominants (Seyfarth, 1977; Henzi & Barrett, 1999; Matheson & Bernstein, 2000; Schino, 2001). Finally, in some species, allogrooming became essential in relationship maintenance (Schino, 2007; Schino et al., 2007; Schino & Aureli, 2008).

Our analysis provides the most comprehensive assessment of the group cohesion hypothesis, as we used the largest sample size to date, included all major taxonomic radiations, and controlled for the effects of phylogenetic inertia. We also clearly differentiated between two social tiers in modular societies and between party vs. community size in species with fission–fusion dynamics. There are relatively few affiliative interactions between units as compared to within units in modular societies (reviewed in Grueter et al., in press-a), so unit size seems to be the appropriate variable here (*contra* Dunbar, 1991; Lehmann et al., 2007). Dunbar (1991) and Lehmann et al. (2007) used band size for species with fission–fusion, but party size is clearly the relevant unit here, as the community as a whole rarely aggregates (Goodall, 1986; White, 1996) and one cannot groom someone who is absent. Our results indicate that the previously reported positive correlation between group size and grooming (Lehmann et al., 2007) may be an artifact of the correlation between terrestriality and group size, partly achieved by excluding the exclusively arboreal Platyrrhini, and of unstable outcomes due to small sample sizes, as well as some errors in data extraction. The group size effect found among Catarrhini (both when we take band and unit size as the relevant unit), produces some continuity with the results of Dunbar (1991) and Lehmann et al. (2007), but this is limited to the Hominoidea. An overall lack of significance of a group size effect and a negative correlation in Platyrrhini clearly show that primates do not seem to use and need grooming to maintain group cohesion.

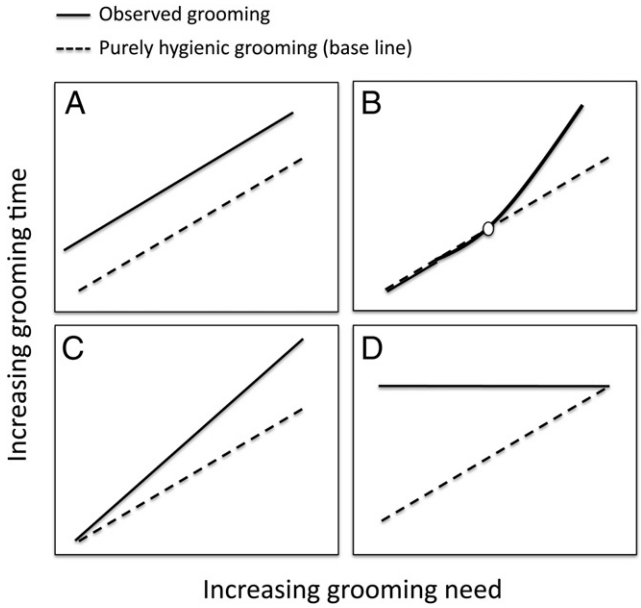



**Fig. 3.** Hypothetical scenarios depicting how the grooming-need hypothesis is compatible with derived, strategic functions of allogrooming. We assume that purely hygienic grooming (dotted line) increases proportionally with increasing grooming need. The observed grooming time (continuous line) includes both hygienic grooming "units" and social grooming "units". The difference between the two lines represents the extra grooming time dedicated to derived, social functions of grooming. Cases a–c explain why we find a strong effect of grooming need (hygiene) on grooming time. (See explanation in text).

The explanatory power of a hypothesis not only depends on whether its predictions are met but also whether there is evidence for the mechanisms it invokes. Dunbar (1991) had suggested two possible ways in which allogrooming could maintain group cohesion with changing group size. One possibility is that, as group size increases, animals must spend proportionately more time grooming because they must allocate grooming time to an increasing number of group members. However, the extreme skew with which most group members groom each other makes this mechanism implausible (e.g. Silk, Seyfarth, & Cheney, 1999; but see Majolo et al., 2009). The second possible mechanism is that, as group size increases, the significance of alliances in countering overcrowding-induced stress may also increase, which should lead animals to invest more allogrooming time in their allies. However, a detailed comparative analysis by Di Bitetti (2000) did not find the expected decrease in grooming diversity with increasing female group size (but see Kudo & Dunbar, 2001). The lack of strong support for the predictions of the group-cohesion model is therefore mirrored by lack of evidence for the two possible mechanisms.

As there was no evidence for the group-cohesion function of allogrooming, this weakens the hypothesis that language evolved as a "cheap" or time-saving form of social grooming that facilitated group cohesion in very large groups of ancestral humans (Dunbar, 1993, 1996; see also Goosen (1987) for a precursor of this idea). Some problems with this gossip theory have previously been noted. First, comparative studies suggest that human conversation is not a more efficient alternative to (chimpanzee) social grooming (Nakamura, 2000). Second, many of the social benefits obtained by language can also be achieved through vocalization in nonhuman primates (Andrew, 1993; see also McComb & Semple, 2005; Seyfarth & Cheney,

1993). Third, humans (with a group size of 150) would not have to spend 42% of total time grooming since the possible relationship between group size (rather than band size) and grooming is asymptotic (Lehmann et al., 2007) and thus there would be an upper limit of ca 20% grooming (Cords, 1993; see also Martins, 1993). Fourth, grooming is a costly behavior that signals commitment to a relationship partner, while language has lower inherent costs and would thus not be a good substitute for grooming (Hauser, Gardner, Goldberg, & Treves, 1993; Power, 1998). The results of the present study thus cast doubt on the cornerstone that the gossip theory is built on, namely an influence of group size on grooming and social cohesion.

## Supplementary Materials

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.evolhumbehav.2012.09.004.

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## References

Agetsuma, N. (1995). Foraging strategies of Yakushima macaques (*Macaca fuscata yakui*). *International Journal of Primatology*, *16*, 595–609.
Aiello, L. C., & Dunbar, R. I. M. (1993). Neocortex size, group size, and the evolution of language. *Current Anthropology*, *34*, 184–193.
Andrew, R. J. (1993). Behavioural constraints on social communication are not likely to prevent the evolution of large social groups in nonhuman primates. *The Behavioral and Brain Sciences*, *16*, 694.
Arnold, C., Matthews, L. J., & Nunn, C. L. (2010). The 10k trees website: A new online resource for primate phylogeny. *Evolutionary Anthropology*, *19*, 114–118.
Aureli, F., Schaffner, C. M., Boesch, C., Bearder, S. K., Call, J., Chapman, C. A., et al. (2008). Fission–fusion dynamics: New research frameworks. *Current Anthropology*, *49*, 627–654.
Aureli, F., Preston, S. D., & de Waal, F. B. M. (1999). Heart rate responses to social interactions in free-moving rhesus macaques (*Macaca mulatta*): A pilot study. *Journalof Comparative Psychology*, *113*, 59–65.
Barrett, L., Henzi, S. P., Weingrill, T., Lycett, J. E., & Hill, R. A. (1999). Market forces predict grooming reciprocity in female baboons. *Proceedingsof the Royal Society B*, *266*, 665–670.
Barrett, L., Gaynor, D., & Henzi, S. P. (2002). A dynamic interaction between aggression and grooming reciprocity among female chacma baboons. *Animal Behaviour*, *63*, 1047–1053.
Bartlett, T. Q. (1999). *Feeding and ranging behavior of the white-handed gibbon* (Hylobateslar) *in KhaoYai National Park, Thailand*. Ph.D. thesis (Washingon University, St. Louis).
Barton, R. (1985). Grooming site preferences in primates and their functional implications. *International Journalof Primatology*, *6*, 519–532.
Boccia, M. L. (1983). A functional analysis of social grooming patterns through direct comparison with self-grooming in rhesus monkeys. *International Journal of Primatology*, *4*, 399–418.
Boccia, M. L., Reite, M., & Laudenslager, M. (1989). On the physiology of grooming in a pigtail macaque. *Physiology and Behavior*, *45*, 667–670.
Boese, G. K. (1974). Social behaviour and ecological considerations of west African baboons (Papiopapio). In R. H. Tuttle (Ed.), *Socioecology and psychology of primates* (pp. 205–230). The Hague: Mouton.
Brain, C., & Bohrmann, R. (1992). Tick infestation of baboons (*Papio ursinus*) in the Nambib Desert. *Journal ofWildlife Diseases*, *28*, 188–191.
Butynski, T. M. (1990). Comparative ecology of blue monkeys (*Cercopithecus mitis*) in high- and low-density subpopulations. *Ecological Monographs*, *60*, 1–26.
Chancellor, R., & Isbell, L. (2009). Female grooming markets in a population of gray-cheeked mangabeys (*Lophocebus albigena*). *Behavioral Ecology*, *20*, 79–86.
Chism, J., & Rogers, W. (2002). Grooming and social cohesion in patas and other guenons. In M. E. Glenn, & M. Cords (Eds.), *The guenons: Diversity and adaptation in African monkeys* (pp. 233–244). New York: Kluwer.
Clutton-Brock, T. H., & Harvey, P. H. (1977). Primate ecology and social organization. In T. H. Clutton-Brock (Ed.), *Primate ecology* (pp. 557–584). London: Academic Press.
Cords, M. (1993). Grooming and language as cohesion mechanisms: Choosing the right data. *The Behavioral and Brain Sciences*, *16*, 697–698.
Delson, E., Terranova, C. J., Jungers, W. L., Sargis, E. J., Jablonski, N. G., & Dechow, P. C. (2000). Body mass in Cercopithecidae (Primates, Mammalia): Estimation and scaling in extinct and extant taxa. *Anthropological Papers of the American Museum of Natural History*, *83*, 1–159.
deRuiter, J. R. (1986). The influence of group size on predator scanning and foraging behaviour of wedge-capped capuchin monkeys (*Cebus olivaceus*). *Behaviour*, *98*, 240–258.
Di Bitetti, M. S. (1997). Evidence for an important social role of allogrooming in a platyrrhine primate. *Animal Behaviour*, *54*, 199–211.
Di Bitetti, M. S. (2000). The distribution of grooming among female primates: Testing hypotheses with the Shannon–Wiener diversity index. *Behaviour*, *137*, 1517–1540.
Di Fiore, A., & Campbell, C. J. (2007). The atelines: Variations in ecology, behavior, and social organization. In C. J. Campbell, A. Fuentes, K. C. MacKinnon, M. Panger, & S. K. Bearder (Eds.), *Primates in perspective* (pp. 155–185). New York: Oxford University Press.
Dunbar, R. I. M. (1991). Functional significance of social grooming in primates. *Folia Primatologica*, *57*, 121–131.
Dunbar, R. I. M. (1993). Coevolution of neocortical size, group size and language in humans. *Behavioraland Brain Sciences*, *16*, 681–694.
Dunbar, R. (1996). *Grooming, gossip and the evolution of language*. London: Faber & Faber.
Dunbar, R. I. M. (2010). The social role of touch in humans and primates: Behavioural function and neurobiological mechanisms. *Neuroscience and Biobehavioral Reviews*, *34*, 260–268.
Dunbar, R. I. M., & Sharman, M. (1984). Is social grooming altruistic? *Zeitschrift für Tierpsychologie*, *64*, 163–173.
Eckstein, R. A., & Hart, B. L. (2000). Grooming and control of fleas in cats. *Applied Animal Behaviour Science*, *68*, 141–150.
Eley, R. M., Strum, S. C., Muchemi, G., & Reid, G. D. F. (1989). Nutrition, body condition, activity patterns, and parasitism of free-ranging troops of olive baboons (*Papio anubis*) in Kenya. *American Journal of Primatology*, *18*, 209–219.
Emmons, L. H. (1984). Geographic variation in densities and diversities of non-flying mammals in Amazonia. *Biotropica*, *16*, 210–222.
Falk, D. (2004). Prelinguistic evolution in early hominins: Whence motherese? *The Behavioral and Brain Sciences*, *27*, 491–541.
Fashing, P. J. (2001). Activity and ranging patterns of guerezas in the Kakamega forest: Intergroup variation and implications for intragroup feeding competition. *International Journal of Primatology*, *22*, 549–577.
Fedurek, P., & Dunbar, R. I. M. (2009). What does mutual grooming tell us about why chimpanzees groom? *Ethology*, *115*, 566–575.
Ford, S. M., & Davis, L. C. (1992). Systematics and body size: Implications for feeding adaptations in New World monkeys. *American Journal of Physical Anthropology*, *88*, 415–468.
Freeland, W. J. (1981). Functional aspects of primate grooming. *Ohio Journal of Science*, *81*, 173–177.
Gamble, C. (1998). Palaeolithic society and the release from proximity: A network approach to intimate relations. *World Archaeol*, *29*, 426–449.
Gittins, S. P., & Raemaekers, J. J. (1980). Siamang, lar and agile gibbons. In D. J. Chivers (Ed.), *Malayan forest primates* (pp. 63–105). New York: Plenum.
Goodall, J. (1986). *The chimpanzees of Gombe*. Cambridge, MA: Belknap Press.
Goosen, C. (1987). Social grooming in primates. In G. Mitchell, & J. Erwin (Eds.), *Comparative primate biology. Vol. 2, part B: Behavior, cognition, and motivation* (pp. 107–131). New York: Alan R. Liss.
Gordon, A. D. (2006). Scaling of size and dimorphism in primates II: Macroevolution. *International Journal of Primatology*, *27*, 63–105.
Grueter CC, Chapais B, Zinner D. (in press-a). Evolution of multilevel societies in nonhuman primates and humans. *International Journal of Primatology*, *33*, 993–1001.
Grueter CC, Li D, Ren B, Li M. (in press-b). Substrate use and postural behavior of free-ranging snub-nosed monkeys (Rhinopithecus bieti) in Yunnan. Integrative Zoology.
Hart, B. L. (1992). Behavioral adaptations to parasites: An ethological approach. *Journal of Parasitology*, *78*, 256–265.
Hauser, M., Gardner, L., Goldberg, T., & Treves, A. (1993). The functions of grooming and language: The present need not reflect the past. *The Behavioral and Brain Sciences*, *16*, 706–707.
Henzi, S. P., & Barrett, L. (1999). The value of grooming to female primates. *Primates*, *40*, 47–59.
Henzi, S. P., & Barrett, L. (2007). Coexistence in female-bonded primate groups. *Advances in the Study of Behavior*, *37*, 43–81.
Henzi, S. P., Lycett, J. E., & Weingrill, T. (1997). Cohort size and the allocation of social effort by female mountain baboons. *Animal Behaviour*, *54*, 1235–1243.
Huang, C., & Li, Y. (2005). How does the white-headed langur (*Trachypithecus leucocephalus*) adapt locomotor behavior to its unique limestone hill habitat? *Primates*, *46*, 261–267.
Huron, D. (2001). Is music an evolutionary adaptation? *Annalsof the New York Academy of Sciences*, *930*, 43–61.
Hutchins, M., & Barash, D. P. (1976). Grooming in primates: Implications for its utilitarian function. *Primates*, *17*, 145–150.
Isler, K., Kirk, E. C., Miller, J. M. A., Albrecht, G. A., Gelvin, B. R., & Martin, R. D. (2008). Endocranial volumes of primate species: Scaling analyses using a comprehensive and reliable data set. *Journal of Human Evolution*, *55*, 967–978.
Karanth, K. U., & Sunquist, M. E. (1992). Population structure, density, and biomass of large herbivores in the tropical forests of Nagarahole, India. *Journal of Tropical Ecology*, *8*, 21–35.
Keverne, E. B., Martensz, N., & Tuite, B. (1989). Beta-endorphin concentrations in cerebrospinal fluid of monkeys are influenced by grooming relationships. *Psychoneuroendocrinology*, *14*, 155–161.
Kirkpatrick, R. C. (1998). Ecology and behavior in snub-nosed and douc langurs. In N. G. Jablosnki (Ed.), *Thenatural history of the doucs and snub-nosed monkeys* (pp. 155–190). Singapore: World Scientific.

Kudo, H., & Dunbar, R. I. M. (2001). Neocortex size and social network size in primates. *Animal Behaviour*, *62*, 711–722.
Kutsukake, N., & Clutton-Brock, T. H. (2006). Social function of allogrooming in cooperatively breeding meerkats. *Animal Behaviour*, *72*, 1059–1068.
Lehmann, J., Korstjens, A. H., & Dunbar, R. I. M. (2007). Group size, grooming and social cohesion in primates. *Animal Behaviour*, *74*, 1617–1629.
Lehmann, T. (1993). Ectoparasites: Direct impacts on host fitness. *Parasitology Today*, *9*, 8–13.
Leinfelder, I., de Vries, H., Deleu, R., & Nelissen, M. (2001). Rank and grooming reciprocity among females in a mixed-sex group of captive hamadryas baboons. *American Journal of Primatology*, *55*, 25–42.
Loudon JE. (2009). The parasite ecology and socioecology of ring-tailed lemurs (Lemur catta) and Verreaux'ssifaka (Propithecus verreauxi) inhabiting the BezaMahafaly Special Reserve. Ph.D. thesis (University of Colorado, Boulder).
Majolo, B., de BortoliVizioli, A., & Schino, G. (2008). Costs and benefits of group living in primates: Group size effects on behaviour and ecology. *Animal Behaviour*, *76*, 1235–1247.
Majolo, B., Ventura, R., Koyama, N. F., Hardie, S. M., Jones, B. M., Knapp, L. A., et al. (2009). Analysing the effects of group size and food competition on Japanese macaque social relationships. *Behaviour*, *146*, 113–137.
Marriott, B. M. (1988). Time budgets of rhesus monkeys (*Macaca mulatta*) in a forest habitat in Nepal and on Cayo Santiago. In J. E. Fa, & C. H. Southwick (Eds.), *Ecology and behavior of food-enhanced primate groups* (pp. 125–149). New York: Alan R. Liss.
Martins, E. P. (1993). Comparative studies, phylogenies and predictions of coevolutionary relationships. *The Behavioral and Brain Sciences*, *16*, 714–716.
Maruhashi, T. (1981). Activity patterns of a troop of Japanese monkeys (*Macaca fuscata yakui*) on Yakushima Island, Japan. *Primates*, *22*, 1–14.
Matheson, M. D., & Bernstein, I. S. (2000). Grooming, social bonding, and agonistic aiding in rhesus monkeys. *American Journal of Primatology*, *51*, 177–186.
McComb, K., & Semple, S. (2005). Coevolution of vocal communication and sociality in primates. *Biology Letters*, *1*, 381–385.
McKenna, J. J. (1978). Biosocial functions of grooming behavior among the common Indian langur monkey (*Presbytis entellus*). *American Journalof Physical Anthropology*, *48*, 503–510.
Mishra, C., & Sinha, A. (2008). A voucher specimen for *Macaca munzala*: Interspecific affinities, evolution, and conservation of a newly discovered primate. *International Journal of Primatology*, *29*, 743–756.
Moore JJ. (1985). *Demography and social behaviour in primates*. Ph.D. thesis (Harvard University, Cambridge).
Mooring, M. S., McKenzie, A. A., & Hart, B. L. (1996). Grooming in impala: Role of oral grooming in removal of ticks and effects of ticks in increasing grooming rate. *Physiology and Behavior*, *59*, 965–971.
Mooring, M. S., Blumstein, D. T., & Stoner, C. J. (2004). The evolution of parasite-defence grooming in ungulates. *Biological Journalof the Linnean Society*, *81*, 17–37.
Nakamura, M. (2000). Is human conversation more efficient than chimpanzee grooming? Comparison of clique sizes. *Human Nature*, *11*, 281–297.
Nelson, H., & Geher, G. (2007). Mutual grooming in human dyadic relationships: An ethological perspective. *Current Psychology*, *26*, 121–140.
Orme D, Freckleton R, Thomas G, Petzoldt T, Fritz S. (2009). *CAIC*: *Comparative analyses using independent contrasts*. R package version 1.0.4-94/r94 (http://R-Forge.R-project.org/projects/caic/).
Perelman, P., Johnson, W. E., Roos, C., Seuánez, H. N., Horvath, J. E., Moreira, M. A. M., et al. (2011). A molecular phylogeny of living primates. *PLoS Genetics*, *7*, e1001342.
Plavcan, J. M., & van Schaik, C. P. (1997). Intrasexual competition and body weight dimorphism in anthropoid primates. *American Journal of Physical Anthropology*, *103*, 37–68.
Port, M., Clough, D., & Kappeler, P. M. (2009). Market effects offset the reciprocation of grooming in free-ranging redfronted lemurs, *Eulemur fulvus rufus*. *Animal Behaviour*, *77*, 29–36.
Power, C. (1998). Old wives' tales: The gossip hypothesis and the reliability of cheap signals. In J. R. Hurford, M. Studdert-Kennedy, & K. Knight (Eds.), *Approaches to the evolution of language* (pp. 111–129). Cambridge: Cambridge University Press.
R Development Core Team. (2010). R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing.
Richard, A. F., Dewar, R. E., Schwartz, M., & Ratsirarson, J. (2000). Mass change, environmental variability and female fertility in wild *Propithecus verreauxi*. *Journal of Human Evolution*, *39*, 381–391.
Rowe N, Myers M. (Eds) 2011. All the world's primates, www.alltheworldsprimates.org. Charlestown RI: Primate Conservation Inc.
Sanchez-Villagra, M. R., Pope, T. R., & Salas, V. (1998). Relation of intergroup variation in allogrooming to group social structure and ectoparasite loads in red howlers (*Alouatta seniculus*). *International Journal of Primatology*, *19*, 473–491.
Saunders, C. D. (1988). Ecological, social and evolutionary aspects of baboon (Papio cynocephalus) grooming behavior. Ph.D. thesis (Cornell University, Ithaca).
Schino, G. (2001). Grooming, competition and social rank among female primates: A meta-analysis. *Animal Behaviour*, *62*, 265–271.
Schino, G. (2007). Grooming and agonistic support: A meta-analysis of primate reciprocal altruism. *Behavioral Ecology*, *18*, 115–120.
Schino, G., & Aureli, F. (2008). Grooming reciprocation among female primates: A meta-analysis. *Biology Letters*, *4*, 9–11.
Schino, G., & Aureli, F. (2010). The relative roles of kinship and reciprocity in explaining primate altruism. *Ecology Letters*, *13*, 45–50.
Schino, G., Polizzi di Sorrentino, E., & Tiddi, B. (2007). Grooming and coalitions in Japanese macaques (*Macaca fuscata*): Partner choice and the time frame of reciprocation. *Journal of Comparative Psychology*, *121*, 181–188.
Seyfarth, R. M. (1977). A model of social grooming among adult female monkeys. *Journal of Theoretical Biology*, *65*, 671–698.
Seyfarth, R. M., & Cheney, D. L. (1993). Grooming is not the only regulator of primate social interactions. *The Behavioral and Brain Sciences*, *16*, 717–718.
Shutt, K., MacLarnon, A., Heistermann, M., & Semple, S. (2007). Grooming in Barbary macaques: Better to give than to receive? *Biology Letters*, *3*, 231–233.
Siex KS. (2003). Effects of population compression on the demography, ecology, and behavior of the Zanzibar red colobus monkey (Procolobus kirkii). Ph.D. thesis (Duke University, Durham).
Silk, J. B., Seyfarth, R. M., & Cheney, D. L. (1999). The structure of social relationships among female savanna baboons in Moremi Reserve, Botswana. *Behaviour*, *136*, 679–703.
Smith, R. J., & Jungers, W. L. (1997). Body mass in comparative primatology. *Journal of Human Evolution*, *32*, 523–559.
Struhsaker, T. T. (1967). Social structure among vervet monkeys (*Cercopithecus aethiops*). *Behaviour*, *29*, 83–121.
Teichroeb, J. A., Saj, T. L., Paterson, J. D., & Sicotte, P. (2003). Effect of group size on activity budgets of *Colobus vellerosus* in Ghana. *International Journal of Primatology*, *24*, 743–758.
Tinbergen, N. (1963). On aims and methods of ethology. *Zeitschrift für Tierpsychologie*, *20*, 410–433.
Ungar, P. S. (1994). Incisor microwear of Sumatran anthropoid primates. *American-Journal of Physical Anthropology*, *94*, 339–363.
White, F. J. (1996). Comparative socio-ecology of *Pan paniscus*. In W. C. McGrew, L. F. Marchant, & T. Nishida (Eds.), *Great ape societies* (pp. 29–41). Cambridge: Cambridge University Press.
Wright, P. C., Arrigo-Nelson, S. J., Hogg, K. L., Bannon, B., Morelli, T. L., Wyatt, J., et al. (2009). Habitat disturbance and seasonal fluctuations of lemur parasites in the rain forest of Ranomafana National Park, Madagascar. In M. A. Huffman, & C. A. Chapman (Eds.), *Primate parasite ecology* (pp. 311–330). New York: Cambridge University Press.
Zamma, K. (2002). Grooming site preferences determined by lice infection among Japanese macaques in Arashiyama. *Primates*, *43*, 41–49.